Embodied AI Challenge

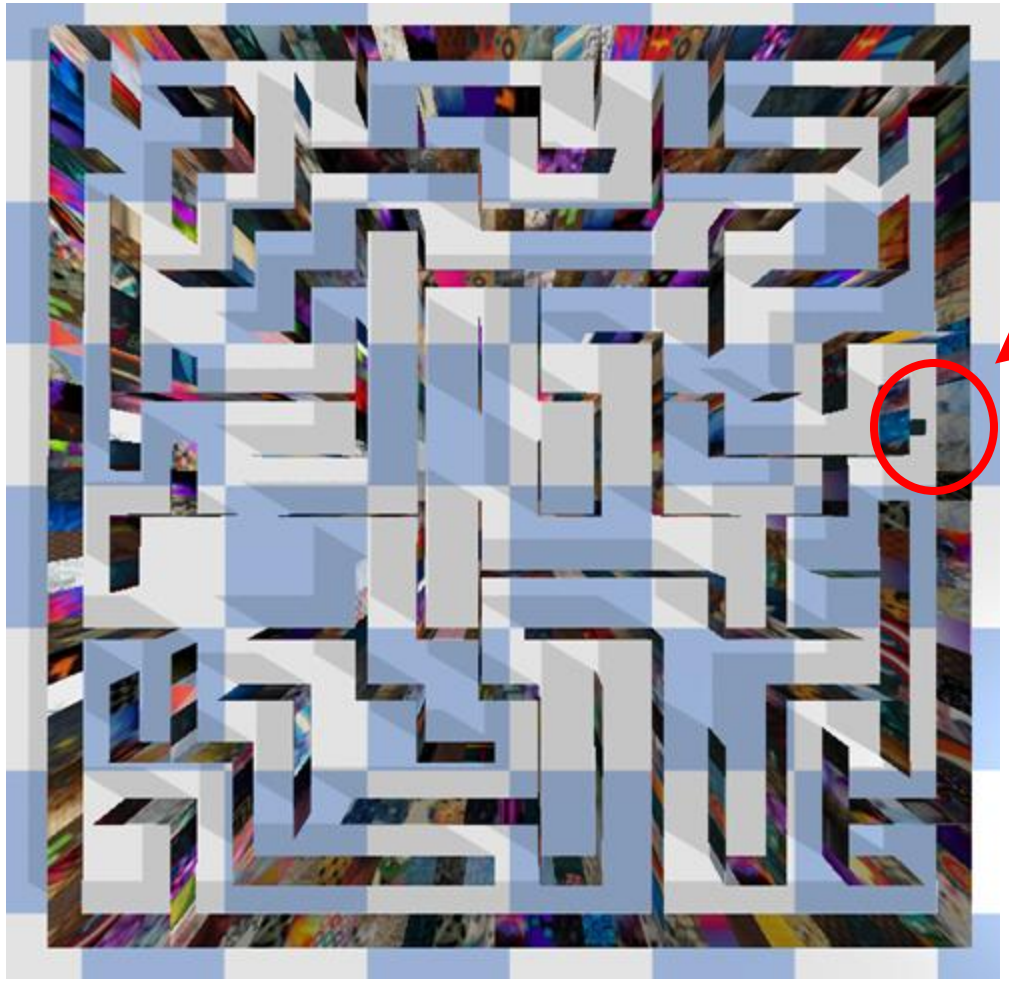
(Visual Navigation Game)

Baseline Solution

Objective

- Given onboard camera views with exploration data, your team should navigate a mobile robot to find goal locations as soon as possible
- It does not require any experience in real-world robotics
- Any methods (fully manual, semi-automatic, fully autonomous) are welcome, but that would affect the final points you receive on this course project
- 1 - 4 students per team
- Detailed evaluation metrics are to be announced in class

Game Environment

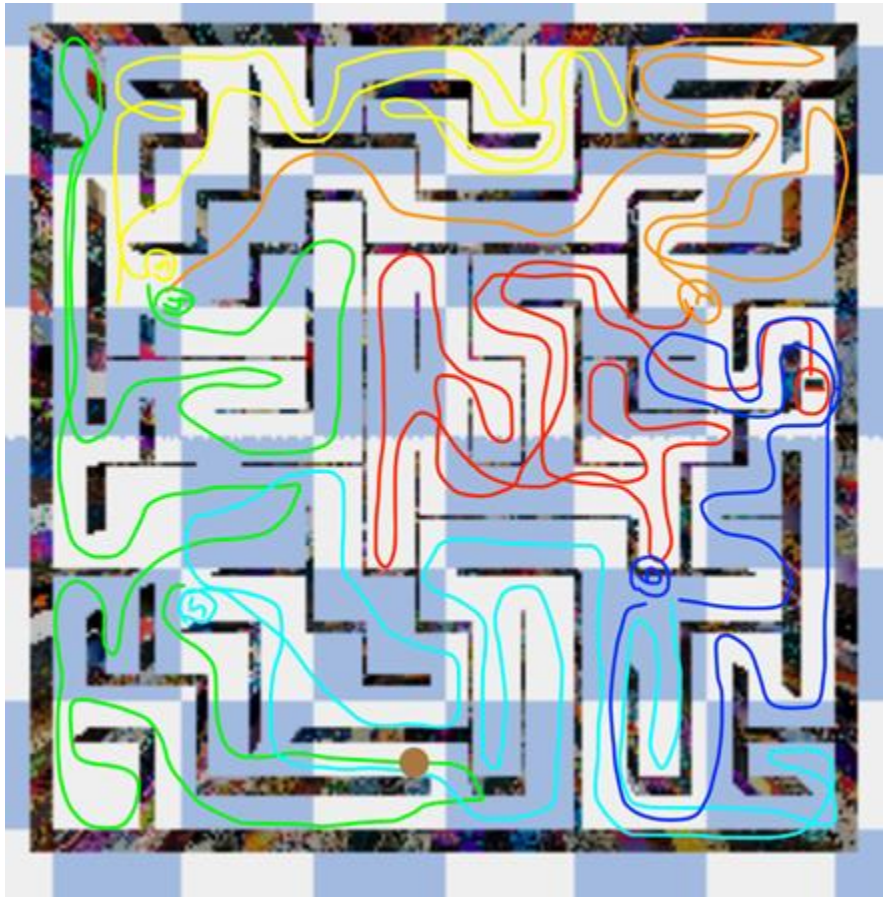


robot starting point

goal information

- Random generated (different maze for competition)
- Maze size: 31 x 31 (demo)
- Texture pattern: 200

Exploration Data



Exploration Path

- Number of Images: 18295
- Image Size: 320x240 (RGB)
- Consider down-sampling in time for initial development

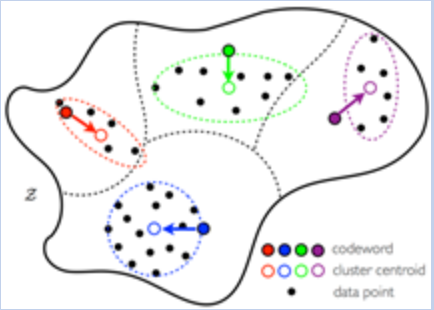
Game Control (Baseline Solution)

- Forward (W)
- Left (A)
- Right (D)
- Backward (S)
- Query Image (Q) user-defined
- Goal Check-in (SPACE)

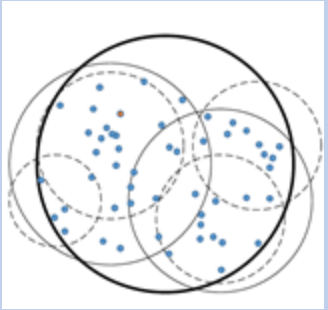
Baseline Solution Pipeline

SIFT Feature

K-Means Cluster



VLAD Descriptor
(Lecture 8)



Ball
Tree

Exploration Images

1. Pre-Nav Compute

2. Navigation Phase

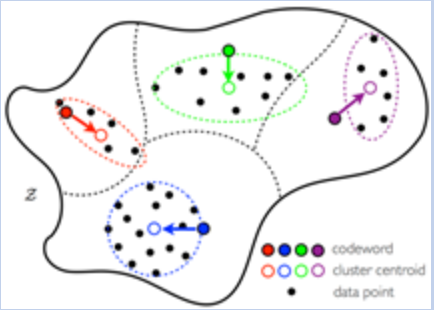
Current View

Nearest Neighbor

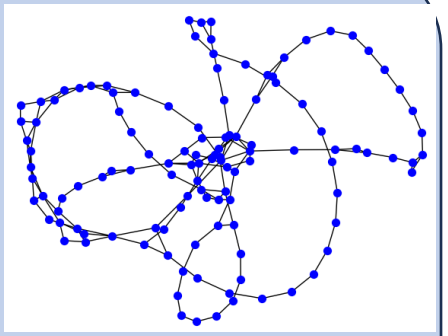
Baseline Solution Pipeline- Updated:

SIFT Features-
Local Image Features

K-Means Cluster-
codebook



VLAD Descriptor



Weighted Graph

Exploration Images

1. Pre-Nav Compute

2. Navigation Phase

Current View

Next Best View
towards Goal

DEMO

- git clone https://github.com/ai4ce/vis_nav_player.git
- cd vis_nav_player
- conda env create -f environment.yaml
- conda activate game
- python baseline.py

Baseline solution- Important Functions:

- *act()*
  - Actions based on keypress by user.
  - Control the player's movement based on player's movement.
  - Upon pressing any other key, the next best view is showed.
  - Saving the keyboard inputs/ actions for evaluation.
- *compute_sift_features()*
  - Using cv2.SIFT_create() to initialize a detector which is used within this function.
  - All the images are accessed using natsorted(natural sorting) and stored within *files* to further access them.
  - For each image the SIFT descriptors are computed and stored into a list which is further converted to an array during function return.
  - SIFT returns the following:
    - Keypoints:
      - Identifies repeatable feature points.
      - Returns location(x,y), scale(size) and orientation(angle).
    - Descriptors:
      - Captures gradient orientations around a keypoint. It is a numerical representation of the feature.
      - Returns a 128-dimensional vector per keypoint. For an image ~ [no of keypoints*128]

Baseline solution- Important Functions:

- *get_VLAD:*
  - Compute 128 dimensional VLAD vectors for each image.
  - For each image:
    - The SIFT descriptor is computed and for those descriptors the cluster labels are predicted.
    - Then, for each cluster available, and the predicted labels, a VLAD vector is computed.
    - The VLAD vector is a sum of all the residual vectors from the centroid of the predicted clusters for the current image.

$$\mathbf{v}_j = \sum_{\mathbf{x}_i \in c_j} (\mathbf{x}_i - \mathbf{c}_j)$$

  - Here, Xi is the descriptor, and Cj is the cluster centroid which form the residual vector. All the residual vectors are added together to form the VLAD vector for that image.
  - The VLAD vector is flattened into a 1xN dimensional vector.
  - We then apply power normalization to each VLAD vector to make them more robust.

Rubrics （old version):

- Report (3 pt)
  - Format: pdf (template)
- Competition (7 pt)
  - Format: Game file in .npy format
  - Rubric:
    1. **Participation** (1 points)
    2. **Achieving the Goal** (2 points)
       - If the position error is below 0.1, then 2, if the error is 0.1 to 0.2, then 1; otherwise, 0.5
    3. **Navigation Time** (2.5 points maximum)
       - Under 1 minute: 2.5 pt
       - Under 2 minutes: 1.5 pt
       - Under 5 minutes: 0.5 pt
    4. **Late Submission Penalty** (-1 points maximum)
       - Submission before 16:00 (0 pts)
       - Submission after 16:00 (-1 pts)
    5. **Methodology & Judge Evaluations** (2.5 points maximum)
       - Judges will evaluate the methodology used and assign points between 0.5 to 2.5, based on the approach and execution. Completely new solution: **2.5**, Modification of the baseline: **1.5**, Baseline solution**: 0.5**
- **Bonus Points [1pt]** - Can be carried to the final competition.

NYU HPC

- https://perfect-resolution-cdd.notion.site/HPC-Cloud-Bursting-Guide-for-ROB-GY-6203-26f88f9b46da800fb038f4afa9fba983